

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Field	Details
Pentester Name (Cybersecurity Professional)	Bushra Alla Awais
Program/Batch	B083C-Networkwalks
Date	23 September 2026
Modules Completed	W2-PM2 (Footprinting with GHDB) & W2-PM5 (Zenmap Scanning)
Client / Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission Secured?	Yes (Authorized Educational Scope & Lab Environment)
Phases Covered	Phase 1: Reconnaissance & Footprinting (GHDB) Phase 2: Scanning & Network Discovery (Zenmap)

1. Liability Disclaimer

I have performed these activities strictly within authorized lab environments and on devices/networks that I own or where testing was conducted passively via public search engine caches. All documentation and materials are for educational and security assessment purposes only. Misuse of scanning tools or unauthorized access against systems without documented owner permission is illegal under applicable cybersecurity legislation.

2. Introduction

This penetration testing report consolidates the practical deliverables completed for **Week 2: Project Module 2 (W2-PM2)** and **Project Module 5 (W2-PM5)**.

The primary objective was to demonstrate the progression from non-intrusive, passive reconnaissance to active network scanning:

- **Phase 1 (W2-PM2 - Footprinting with GHDB):** Utilizing Google dorks from the Exploit-DB Google Hacking Database to uncover sensitive assets, unauthenticated IoT devices (such as live camera feeds), and open document indexes directly from Google's indexed cache without generating traffic to the target servers.
- **Phase 2 (W2-PM5 - Network Scanning with Zenmap):** Employing Zenmap (Nmap GUI) on a Windows 10 host to execute ping sweeps (`nmap -sn`), discover active hosts within the local subnet (10.120.115.0/24), extract Layer 2 MAC addresses, and generate an interactive topology map.

3. Tools & Technologies Used

Tool / Technology	Platform	Purpose in Engagement
Google Hacking Database (GHDB)	Web (exploit-db.com)	Sourcing advanced search dorks to identify exposed systems and directory indices.
Google Search Engine	Web Browser	Executing crafted queries to footprint publicly leaked assets passively.
Windows Command Prompt (<code>ipconfig /all</code>)	Windows 11/10	Identifying host adapter IPv4/IPv6 configuration, default gateway, and local physical MAC address.
Zenmap (Nmap 7.991)	Windows	Performing ARP/ICMP ping sweeps, active host enumeration, and topology graphing.
Npcap Packet Driver	Windows Driver	Packet capture and raw packet transmission library supporting Nmap scanning on Windows.

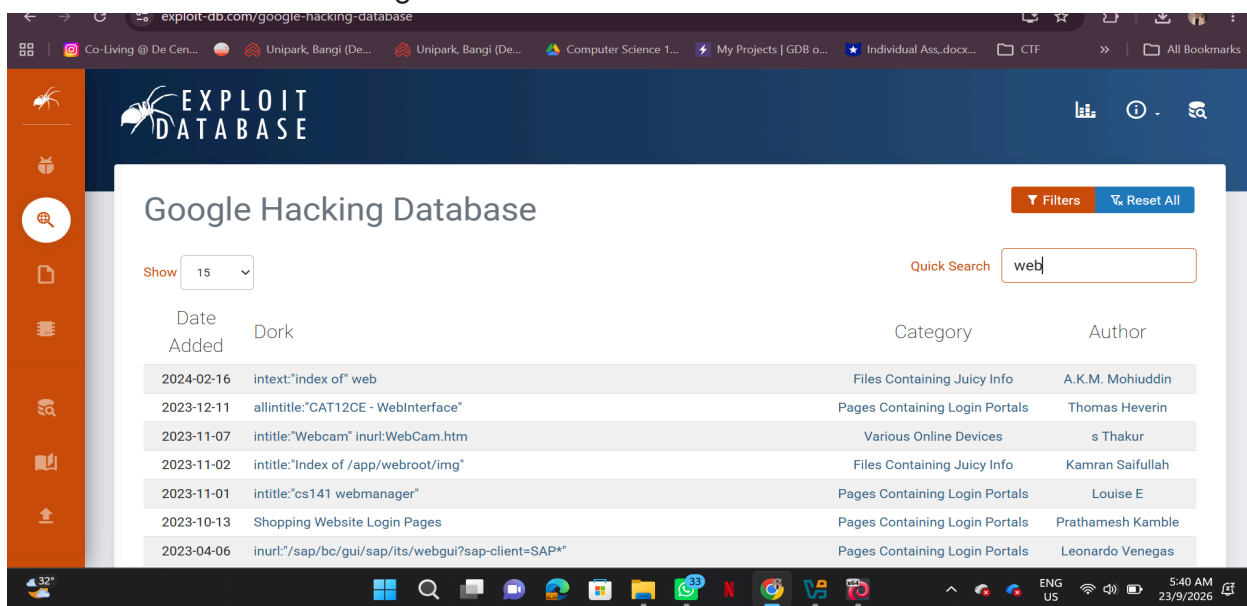
4. Activities Performed

4.1 Phase 1: Passive Footprinting with GHDB (W2-PM2)

The GHDB repository hosted on Exploit-DB was accessed to extract dorks categorised under *Various Online Devices* and *Pages Containing Login Portals*. Because search requests are

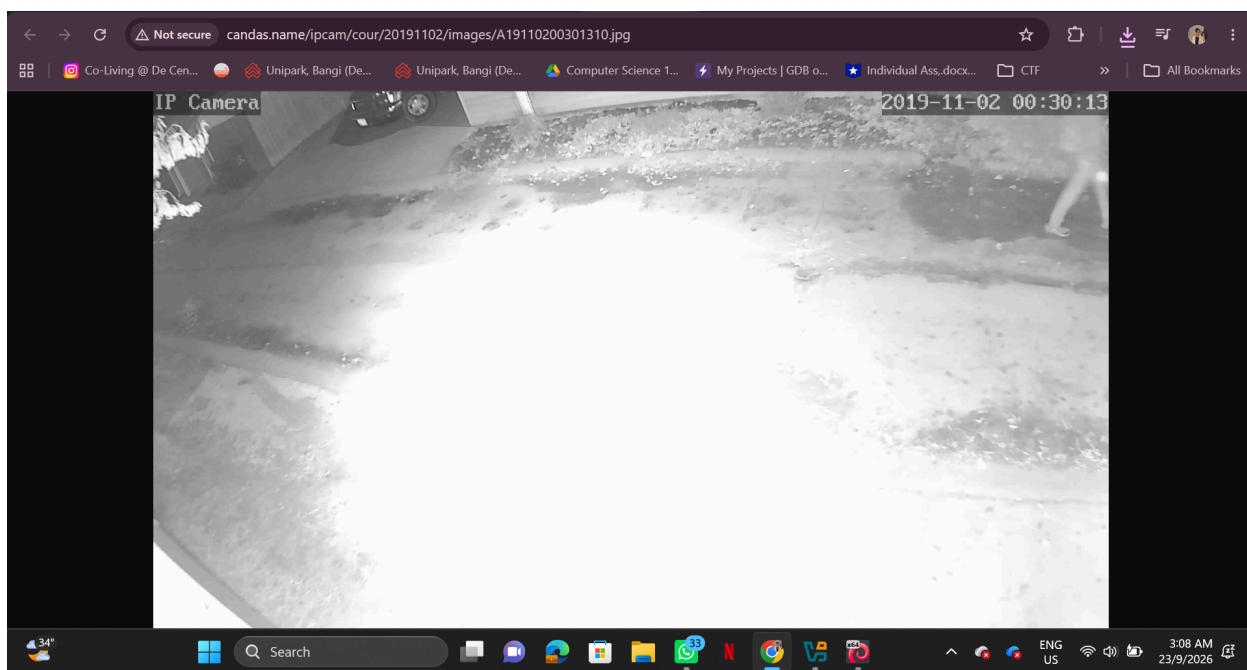
resolved directly by Google's web scrapers, no direct connection was initiated against target infrastructures during discovery.

1. **Locating Exposed Camera Infrastructure:** Queries filtering for webcamXP web servers on non-standard ports (8080) were executed. Multiple web interfaces were identified where live surveillance feeds were exposed without requiring authentication.
2. **Locating Open Directory Document Listings:** Directory-traversal dorks containing `intitle:index.of` were queried alongside `mathematics pdf` parameters to identify unsecured file servers hosting academic and technical literature.

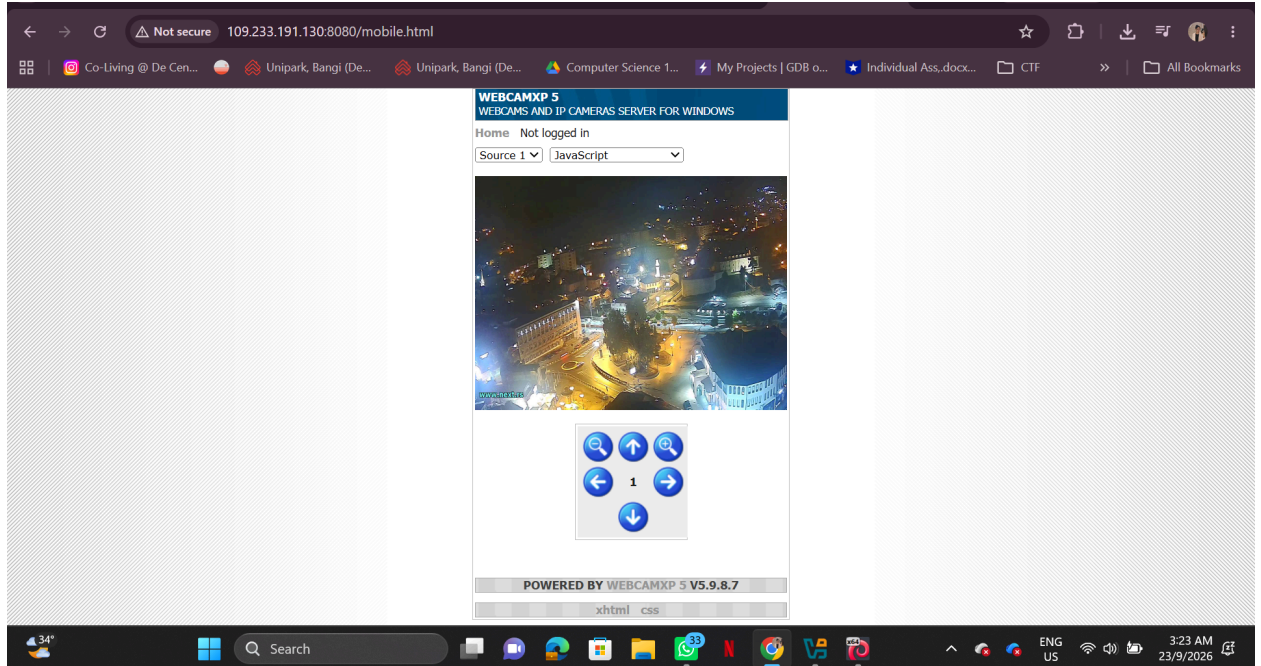
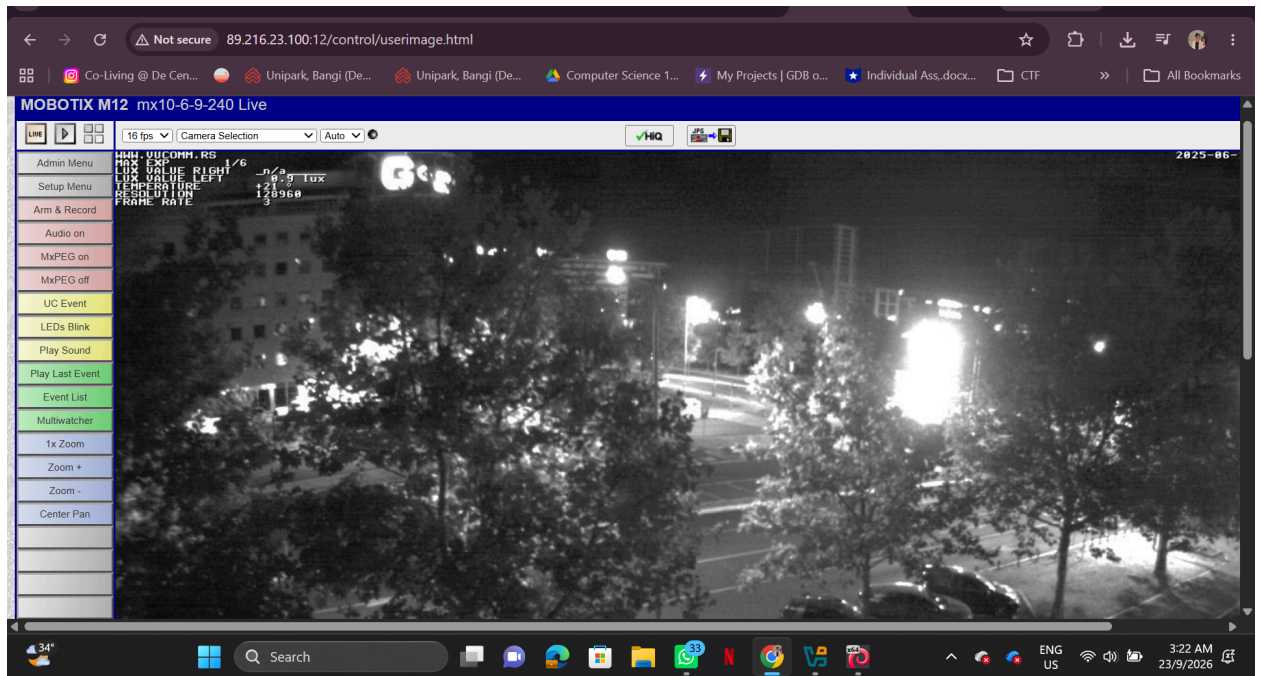


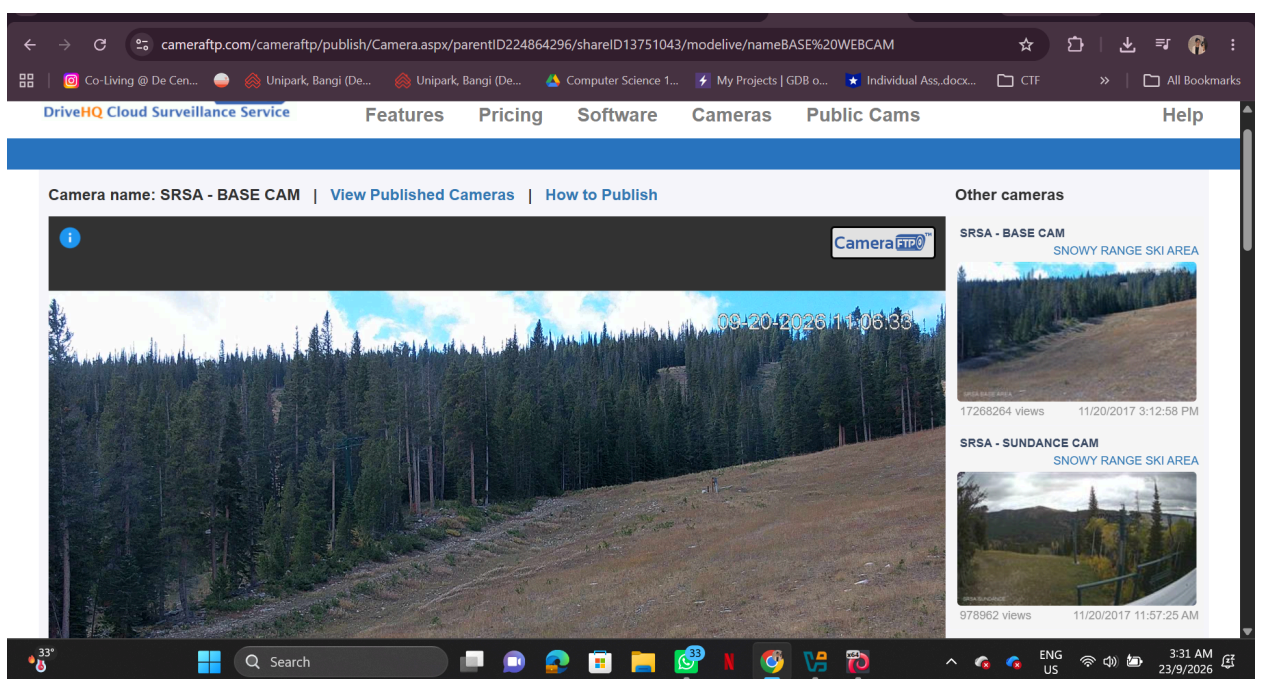
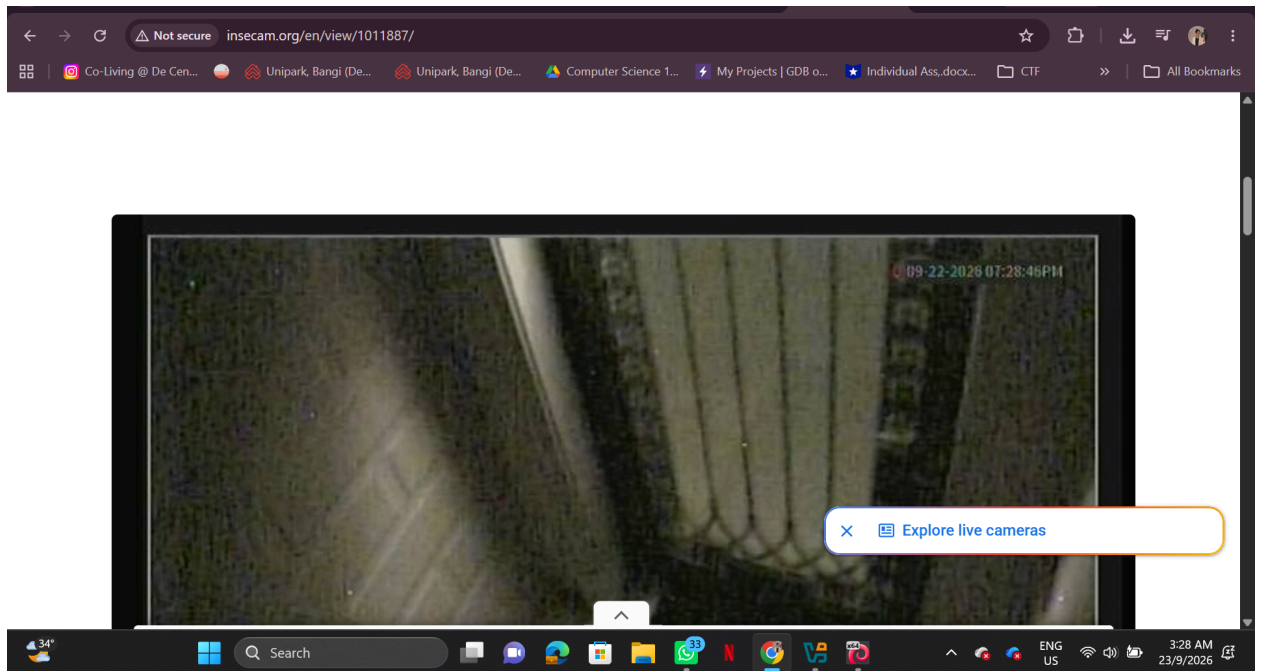
The screenshot shows the Exploit Database website interface. The main heading is "Google Hacking Database". Below it, there is a search bar with the text "web" entered. The results are displayed in a table with columns: Date Added, Dork, Category, and Author.

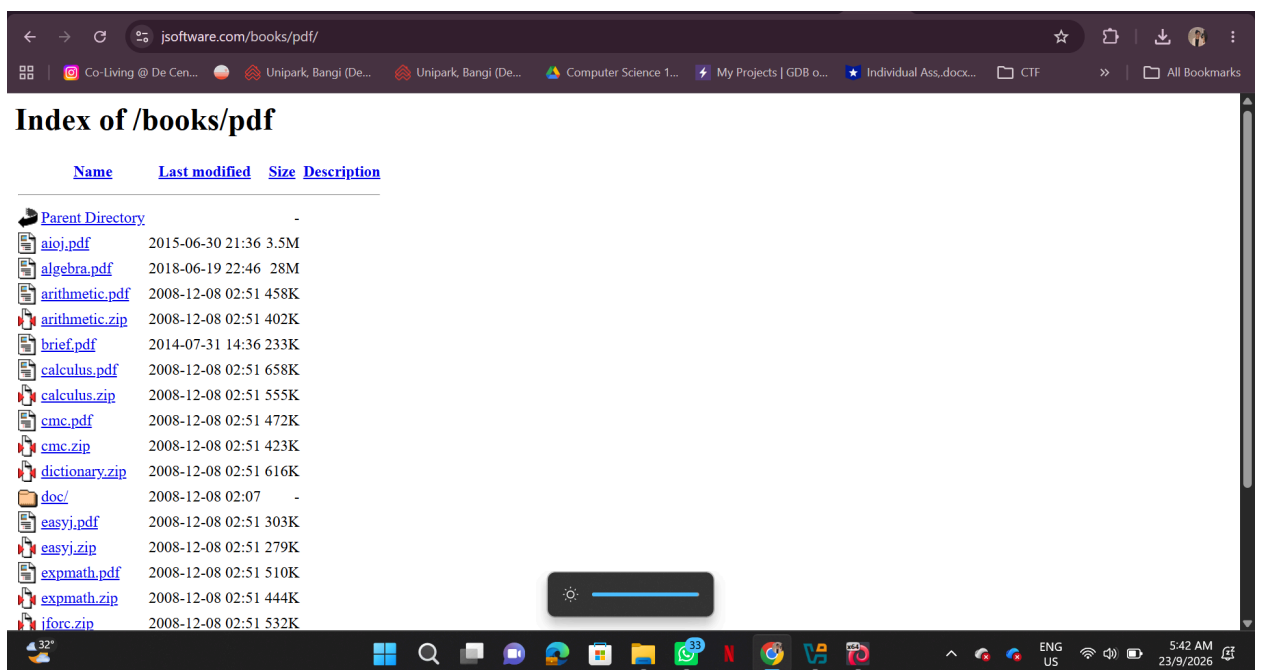
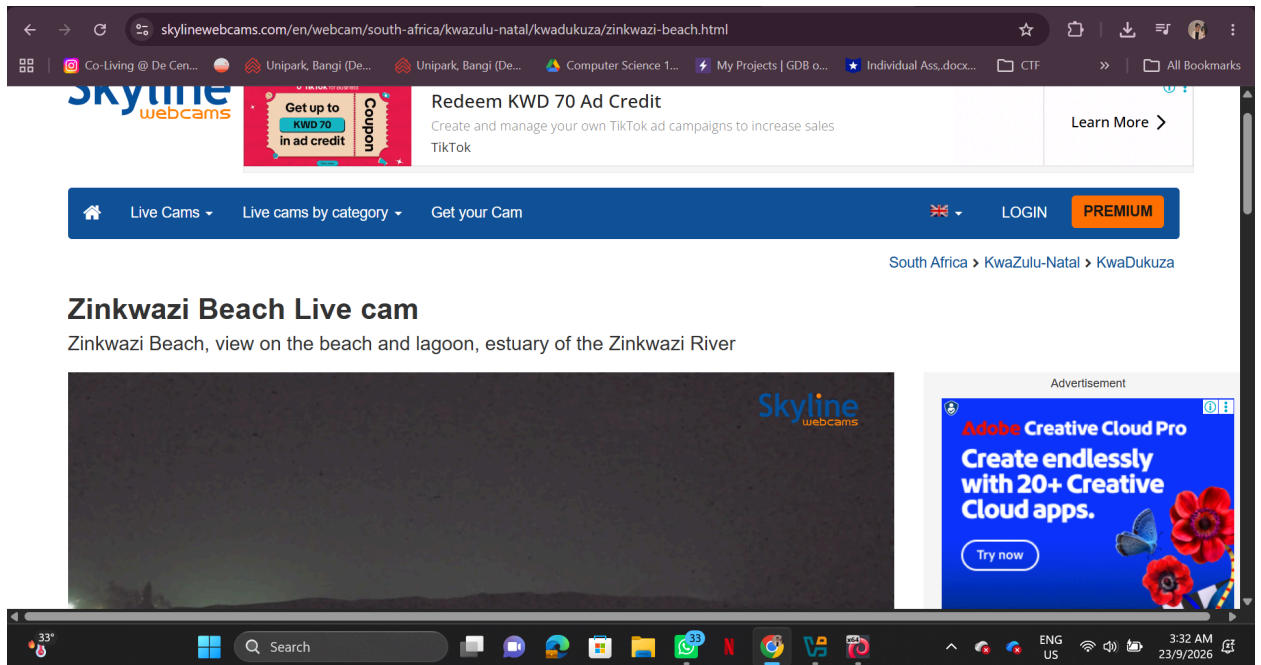
Date Added	Dork	Category	Author
2024-02-16	<code>intext:"index of" web</code>	Files Containing Juicy Info	A.K.M. Mohiuddin
2023-12-11	<code>allintitle:"CAT12CE - WebInterface"</code>	Pages Containing Login Portals	Thomas Heverin
2023-11-07	<code>intitle:"Webcam" inurl:WebCam.htm</code>	Various Online Devices	s Thakur
2023-11-02	<code>intitle:"Index of /app/webroot/img"</code>	Files Containing Juicy Info	Kamran Saifullah
2023-11-01	<code>intitle:"cs141 webmanager"</code>	Pages Containing Login Portals	Louise E
2023-10-13	Shopping Website Login Pages	Pages Containing Login Portals	Prathamesh Kamble
2023-04-06	<code>inurl:"/sap/bc/gui/sap/its/webgui?sap-client=SAP"</code>	Pages Containing Login Portals	Leonardo Venegas



The screenshot shows a live IP camera feed. The address bar indicates the URL is `candas.name/ipcam/cour/20191102/images/A19110200301310.jpg`. The video feed shows a street scene with a car and a person. The timestamp in the top right corner of the video is "2019-11-02 00:30:13".







Summary of Findings: Exposed IoT Devices & Open Repositories

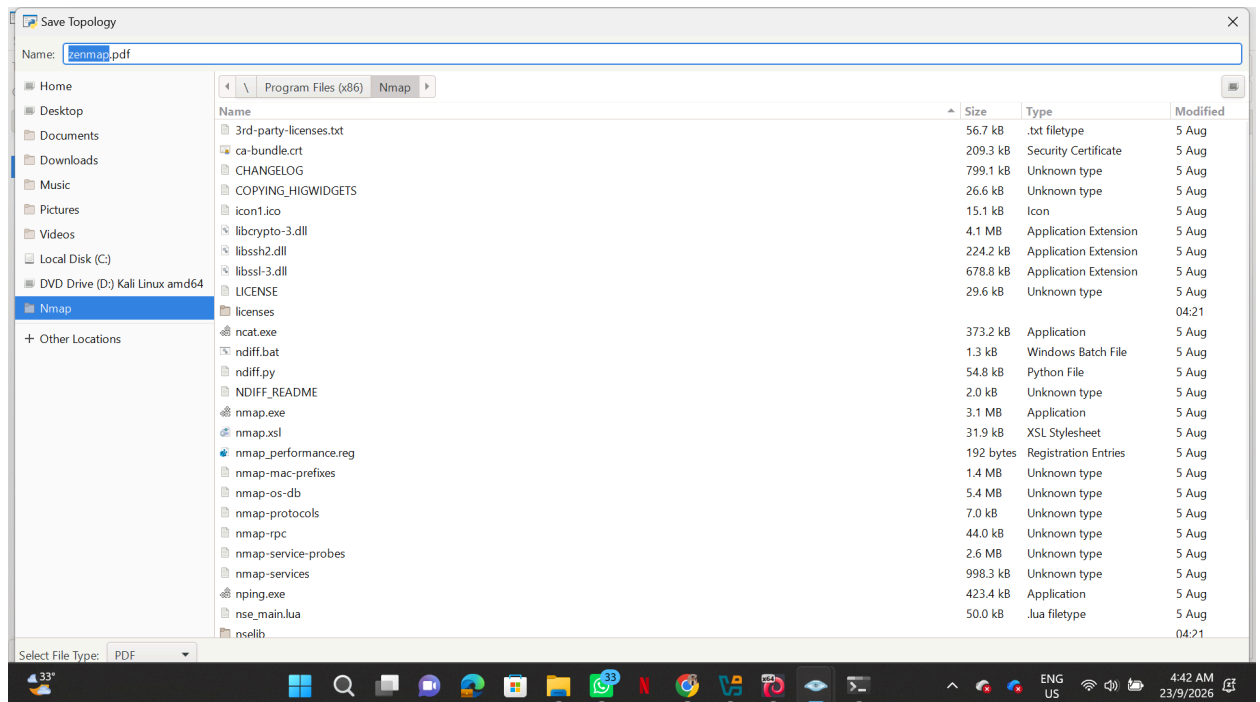
- Camera Systems (Task 1):** Identified live surveillance installations running webcamXP 5 across multiple public endpoints (e.g., http://122.116.41.8:8080/, http://99.132.182.35:8080/, http://104.177.153.145:8080/) displaying real-time video streams and pan/tilt controls with zero authentication barriers.
- Document Indexes (Task 2):** Uncovered open parent directories hosting educational

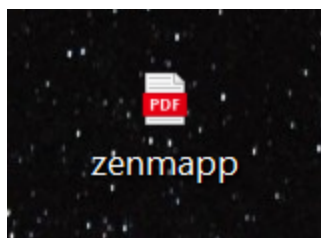
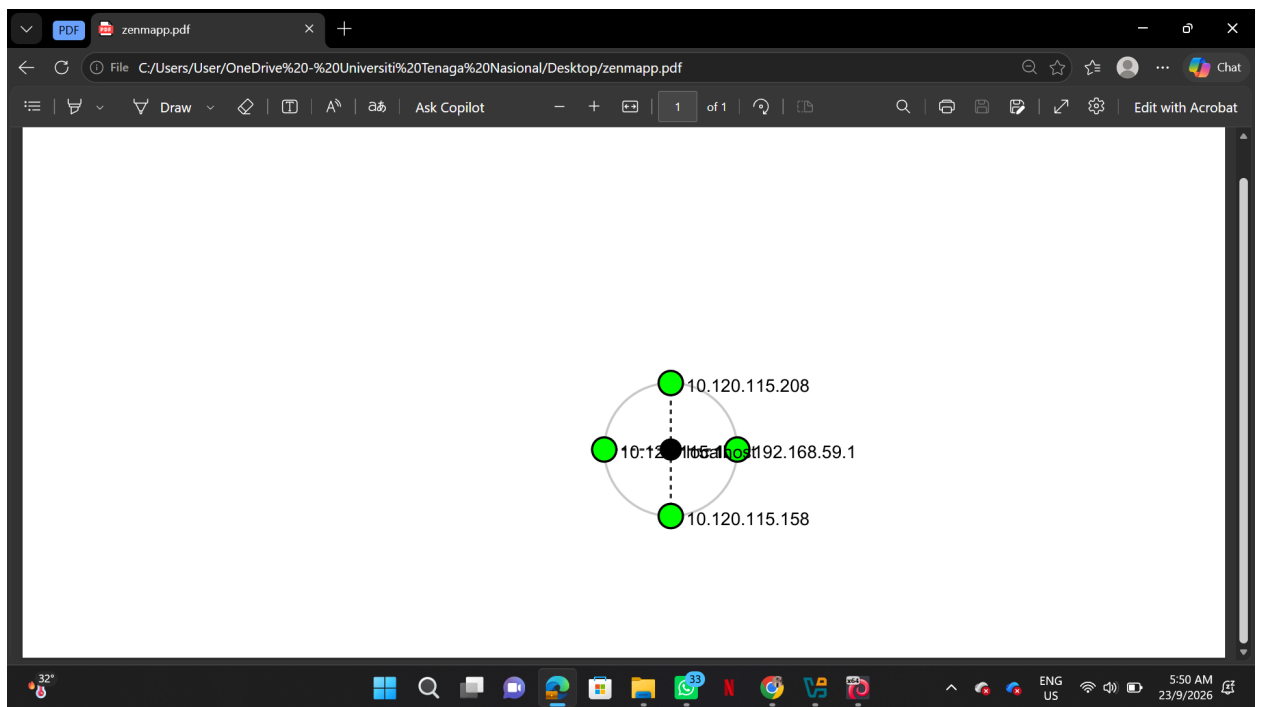
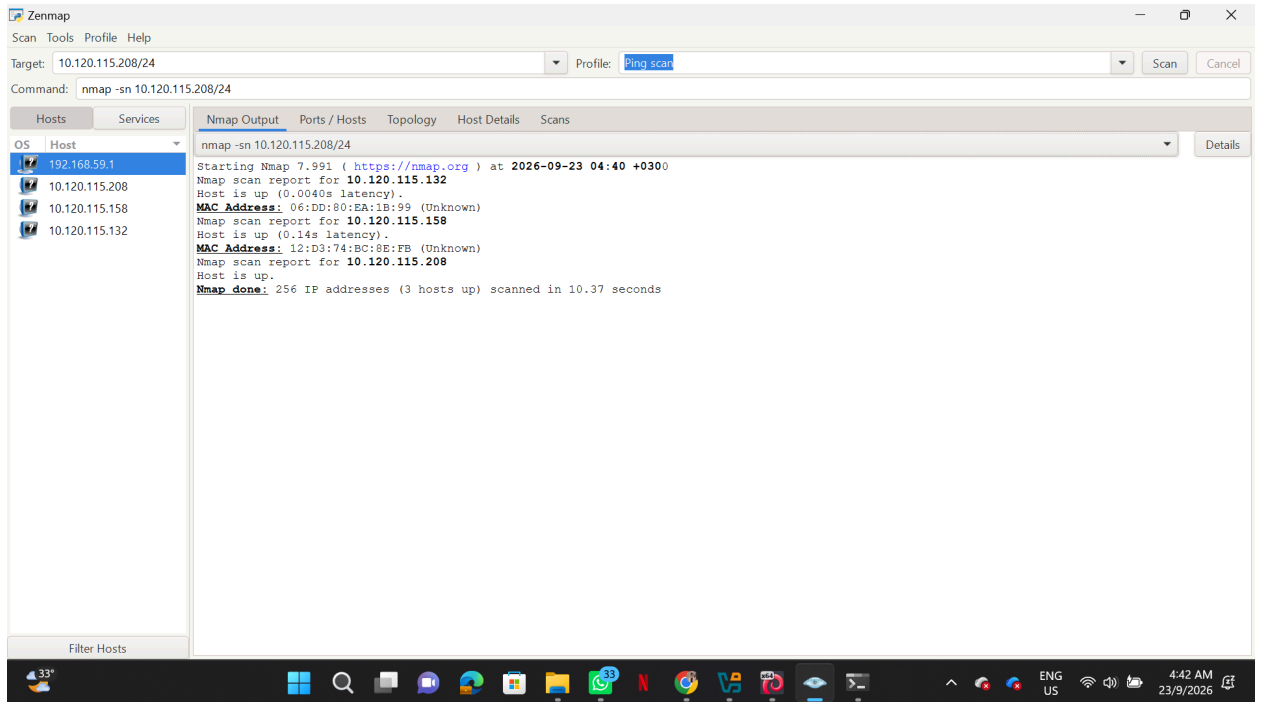
PDFs (e.g., Skyline University College index at https://www.skylineuniversity.ac.ae/pdf/math/ and CARMA Newcastle repository at https://carma.newcastle.edu.au/brailey/Lecture_Notes/) revealing internal file trees and unindexed learning materials.

4.2 Phase 2: Active Network Scanning with Zenmap (W2-PM5)

Active host enumeration was executed locally using Zenmap on Windows.

1. **Local Subnet Identification:** Command Prompt was utilized to run `ipconfig /all`. The active interface (Wireless LAN adapter Wi-Fi, Intel Dual Band Wireless-AC 3165) identified the host IP address as 10.120.115.208, the subnet mask as 255.255.255.0 (/24), the default gateway as 10.120.115.132, and the host's physical MAC address as AC-2B-6E-74-9C-7B.
2. **Ping Scan Execution:** A ping sweep was initiated across the subnet CIDR block via the Zenmap interface using the Ping scan profile (`nmap -sn 10.120.115.208/24`).
3. **MAC & Hardware Enumeration:** Layer 2 ARP requests quickly determined the hardware MAC addresses for remote active devices on the local wireless subnet.
4. **Topology Map Export:** The network topology was generated in Zenmap and exported as a PDF file (`zenmap.pdf`).





Live Host Inventory (10.120.115.0/24)

Host IP Address	State	MAC Address	Interface / Role
10.120.115.132	Up	06:DD:80:EA:1B:99	Default Gateway / DHCP & DNS Server
10.120.115.158	Up	12:D3:74:BC:8E:FB	Active Network Host
10.120.115.208	Up	AC:2B:6E:74:9C:7B (via <i>ipconfig</i>)	Scanning Host (localhost) / Intel Wi-Fi Adapter

Total hosts identified as active: **3 hosts up.**

5. Risk Analysis & Impact Assessment

#	Observation / Finding	Technical Impact	Risk Level
1	Unauthenticated IP Video Streams Exposed via GHDB	Unrestricted access to private spaces, physical surveillance bypass, and exposed administrative login endpoints.	Critical
2	Exposed Directory Browsing (Index of /)	Uncontrolled data leakage, revelation of file path schemas, and unauthenticated exfiltration of proprietary documents.	Medium
3	Internal Host Discovery via Ping Sweep	Attackers connected to an internal WLAN can map the active IP landscape, identify the default	Medium

		gateway, and target neighboring nodes without full port scanning.	
4	Hardware MAC Address Disclosure via ARP	ARP scanning across the local layer reveals hardware signatures and device presence, aiding targeted Layer 2 attacks like ARP spoofing.	Low

6. Recommendations & Mitigation Strategies

1. **Implement Restrictive Robots & Header Directives:** Enforce X-Robots-Tag: noindex, nofollow headers and configure robots.txt to instruct web search crawlers against cataloging private administrative panels and internal file trees.
2. **Disable Directory Browsing:** Modify web server configurations (e.g., Options -Indexes in Apache or removing directory browsing modules in Nginx/IIS) to prevent open file tree enumeration when default index files are missing.
3. **Enforce IoT Authentication Baselines:** Isolate surveillance systems into dedicated, non-routable management VLANs and mandate multi-factor authentication (MFA) or robust password controls on web interfaces before allowing streams to initialize.
4. **Wireless Client Isolation:** Enable client/AP isolation on the wireless router/gateway (10.120.115.132) to prevent wireless endpoints from sending direct ARP/unicast probes to one another on the same subnet.
5. **Periodic Defensive Auditing:** Security personnel should periodically execute self-targeted Google dorks against company domains and conduct scheduled internal Zenmap/Nmap sweeps to detect unauthorized rogue devices.

7. Conclusion

Week 2 practical modules demonstrated the critical transition between passive open-source reconnaissance and active internal network enumeration.

Through **W2-PM2**, I evaluated how indexing engines can be leveraged to locate unprotected IoT equipment and sensitive documents without ever contacting the target host. Through **W2-PM5**, I demonstrated the mechanics of host discovery and network mapping using Zenmap on my own network, identifying active devices, MAC addresses, and the default gateway in seconds. Both exercises reinforce that minimizing internet-facing exposure and maintaining strict internal network segmentation are vital components of defensive architecture.

8. Evidence & Artifacts Log

Figure Ref	Artifact Description	File / Command Reference
Fig 1.1	Exploit-DB GHDB Category Navigation	Browser: [exploit-db.com/google-hacking-database] (https://exploit-db.com/google-hacking-database)
Fig 1.2	Google Dork Search Output for Cameras	Dork: intitle:"webcamXP" inurl:8080
Fig 1.3	Live Unauthenticated WebcamXP Panel	URL: http://122.116.41.8:8080/
Fig 1.4	Open Mathematics Directory Index	Dork: intitle:index.of "parent directory" mathematics pdf
Fig 2.1	Windows ipconfig /all Configuration	Screenshot: Screenshot (524).jpg
Fig 2.2	Zenmap Discovery Ping Scan Execution	Screenshot: Screenshot (525).png
Fig 2.3	Exporting Topology Graphic to PDF	Screenshot: Screenshot (526).png

Author Information

Bushra Alla Awais

Cybersecurity Professional | Batch B082

Program: Cybersecurity Program at Networkwalks